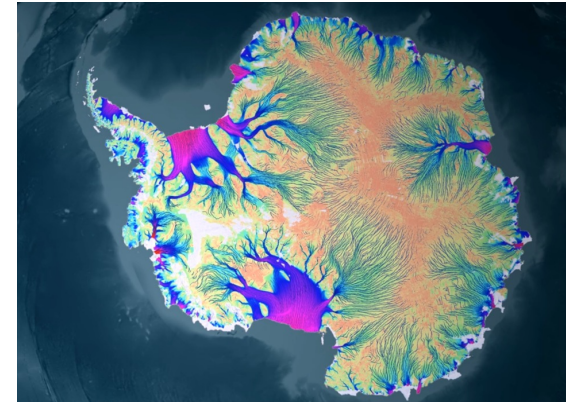
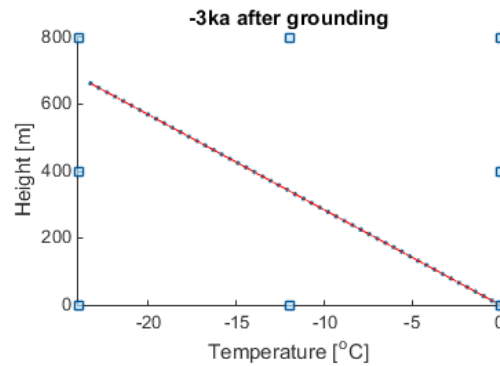
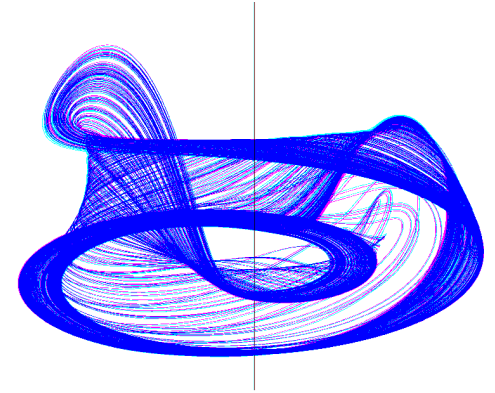
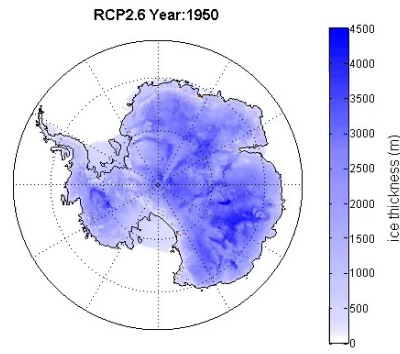


Glaciology

EESCGU4220

3 credits
Fall 2025

Instructor:
Jonny Kingslake



Glaciology

EESCGU4220

3 credits
Fall 2025

Class sessions

Tuesdays: 2:40 pm ET, Schermerhorn ext. 555

Thursdays: 2:40 pm ET, Schermerhorn ext. 555

Office hours

10 am, Schermerhorn Ext. 561

Instructor:

Jonny Kingslake



Lecture 1:

Introduction to Glaciers and Ice Sheets

- Introductions
- The course
 - Grading
 - Assignments
 - Expectations
 - Textbooks
- Glaciers and ice sheets
- Overview of course topics

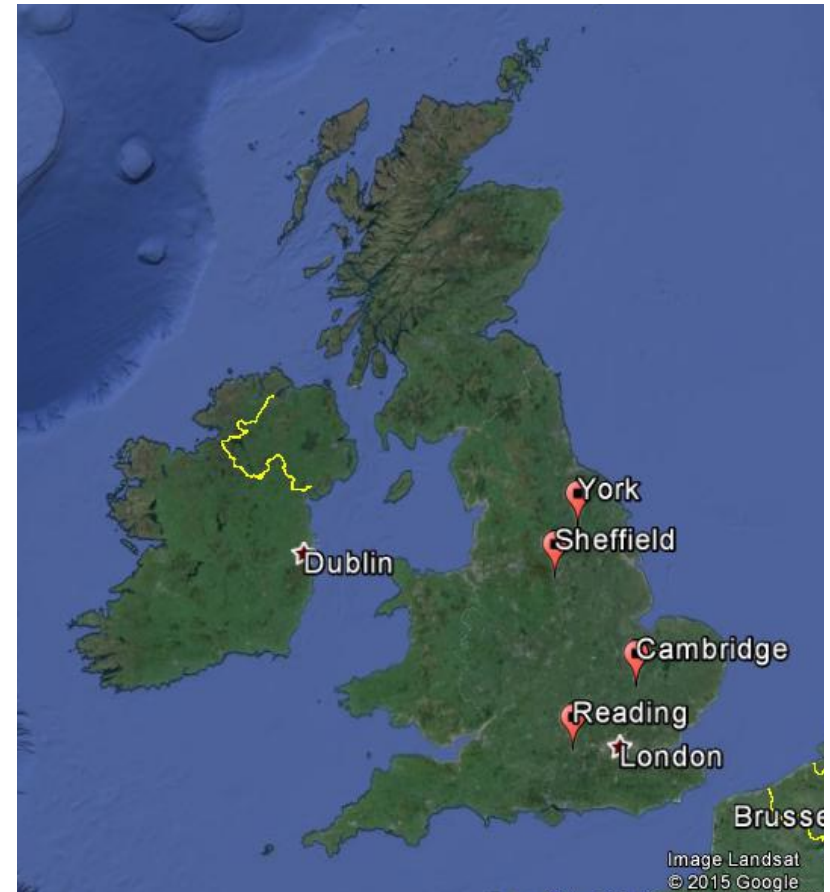
Bio

Undergraduate degree in physics,
The University of York.

PhD in mathematical glaciology,
The University of Sheffield:
'Modelling ice-dammed lake drainage'.

Glacier geophysicist
British Antarctic Survey, Cambridge.

Associate Professor
Lamont-Doherty Earth Observatory,
Columbia University



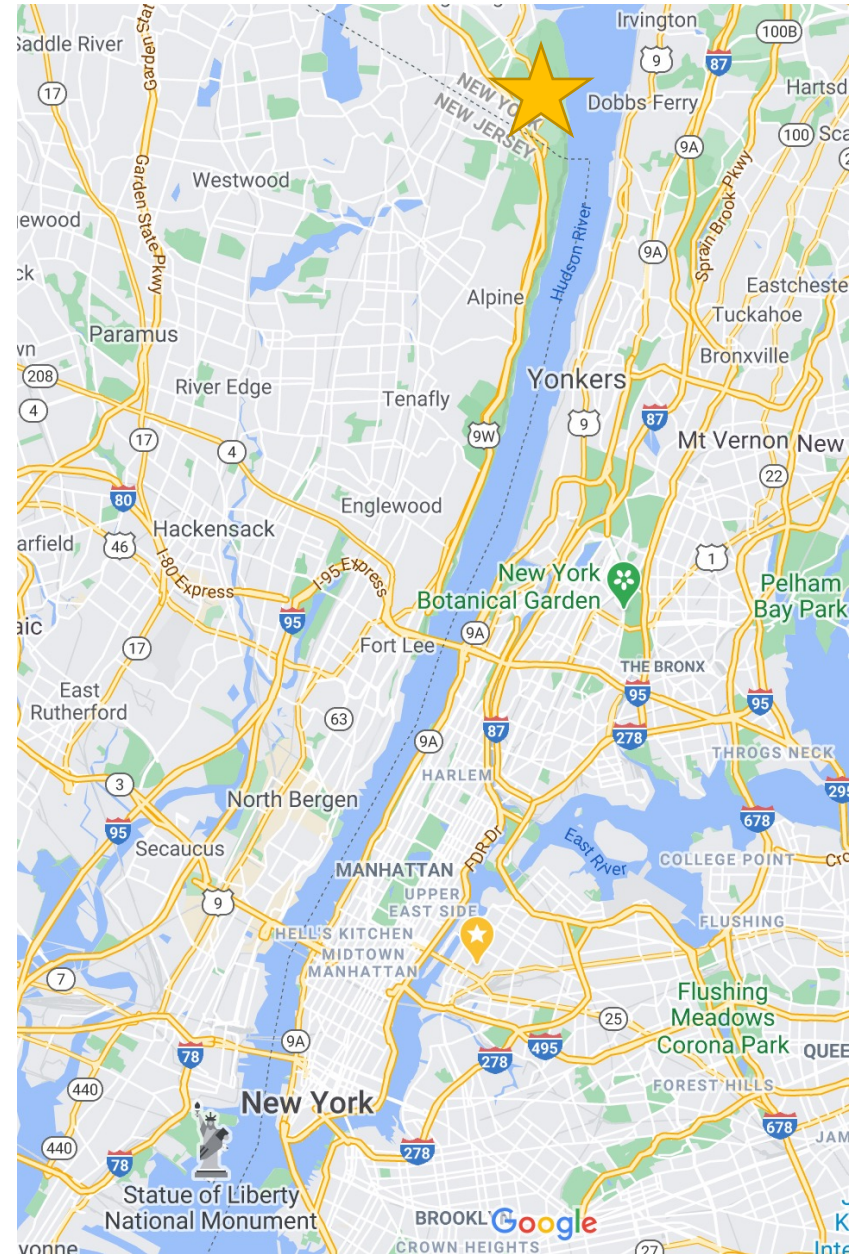
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Lamont-Doherty Earth Observatory,
Columbia University



Introductions

- Name

Jonny

- Current main scientific interest

Flow of water and ice in glaciers and ice sheets
(mostly Antarctica).

- Reasons for being here

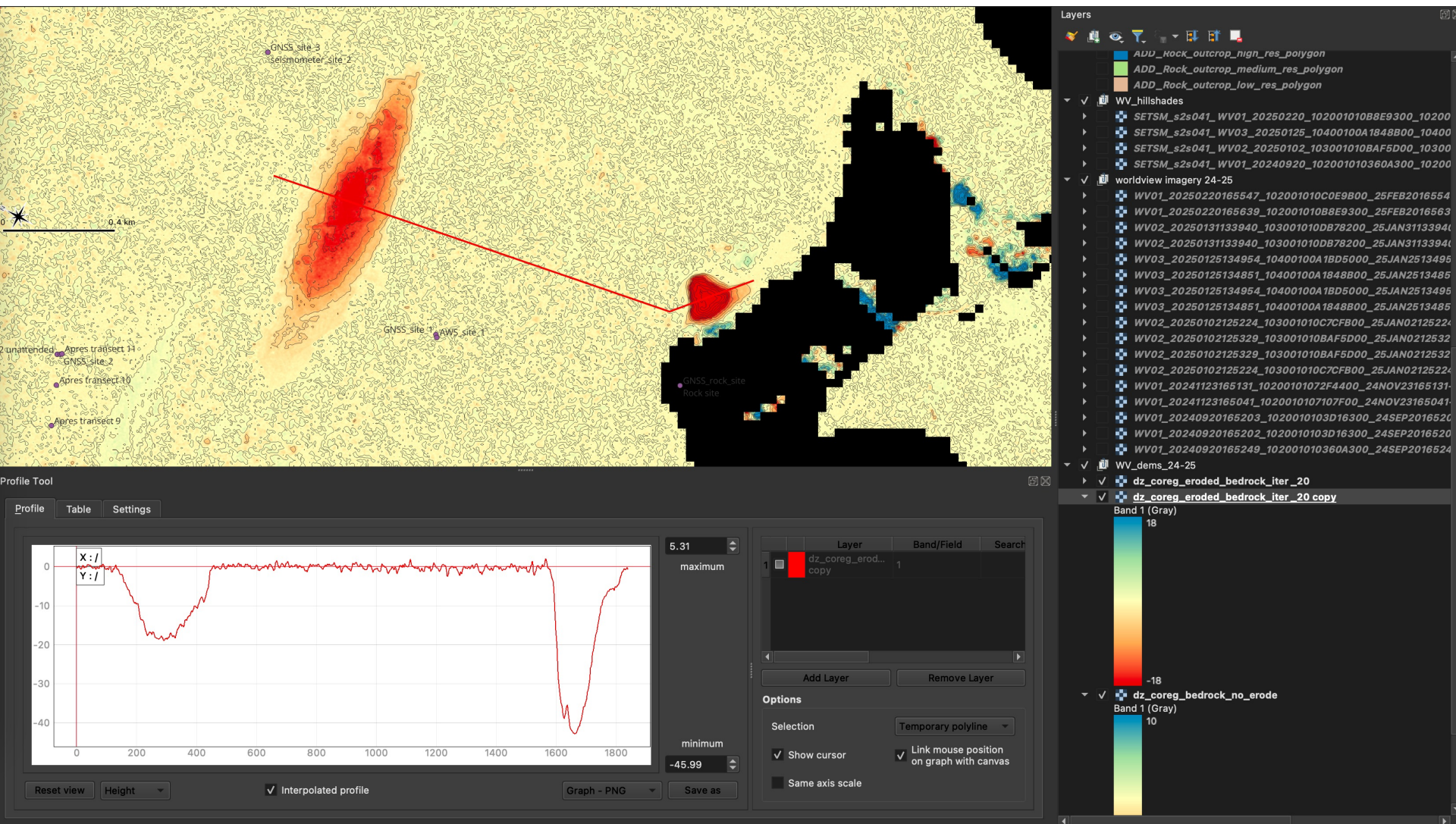
Because I enjoy talking about glaciology
(also its my job).

- One thing that you want to better understand this semester (with a figure if you like).

Lakes on Flask Glacier, Antarctica.

Flask Glacier, Antarctica





Madeleine Yao

- Interested in: subglacial hydrology / radar
- Hoping to: start and continue studying glaciers
- Learn: physics of ice flow, modeling techniques



Name: Mackenzie Blanus

Scientific Interest: High salinity shelf water formation and transport in Terra Nova Bay, Antarctica, as inferred from oceanographic moorings

Why I'm here: To learn about glaciers and their connection to oceans

Better understand: How glacial / ice sheet melting will affect the global ocean circulation. How to analyze glaciers / ice via satellites. Very interested in SAR data.

I saw glaciers for the first time this summer!!!



Jack Bhalla

Current Main Scientific Interest:

My main scientific interest is probably my major, computer science. However, seeing as that is a relatively broad term, I would say specifically the study and applications of computer vision.

Reason For Being Here:

I primarily selected this course to serve as one of my General Technical Electives. However, that is not to say it wasn't the most interesting available course. I think I am primarily interested in this subject because I am an avid hiker and backpacker, so glaciers and their impact on the formation of terrain has always been something of interest to me.

Better Understanding:

Primarily, I am hoping to learn more about applications of data science and numerical modeling, as well as any climate-oriented responses of ice sheets and glaciers.

Intro

Name: Miranda Dawson

Current Scientific Interest: Glacial isostatic adjustment and sea level rise

Reason for Being Here: Glaciers are super cool!!

One thing to better understand this semester:

Excited to learn more about modeling glaciers and data analysis/math of glaciology

The course

Tuesdays: lectures (and discussion).

Tuesdays: 2:40 pm
555 Schermerhorn
Extension

- 1hr 15 mins
- Introduce and discuss the most important processes
- Observations
 - Satellite
 - Geophysics
 - Lab experiments
- Mathematical models
 - Fundamental physics (balance of energy, mass and stress).
 - Partial differential equations
 - These will be very simplified
- Examine models analytically

Emphasis on discussion and understanding processes!

Format:

Thursdays: practical sessions.

Practicals:

- 1hr 15 mins

Thursdays: 1:10pm 555 Schermerhorn Extension

Format:

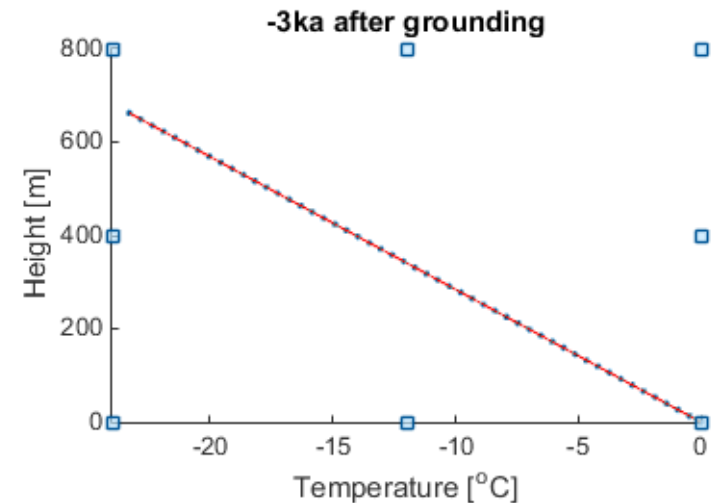
Thursdays: practical sessions.

Thursdays: 1:10pm 555
Schermerhorn Extension

Practicals:

- 1hr 15 mins
- Analyzing mathematical models numerically
 - a simple sets of equations for ice flow
 - heat flow through ice sheets

$$q = u_b H + \frac{2A}{n+2} \rho g H^{n+2}$$



Grading

Assignments (3): 40%

Completing work in practicals

Final paper/project: 40%

A written paper on a glaciological topic, going beyond what we discuss in class, or

A report on a computer-based project
(could extend a model or data analysis exercise from class).

(0.5-page plan by Nov 7th and submit by Dec 19th)

Class participation: 20%

Discussion/contribution in lectures, practicals, or office hours

Grade	lower bound
A+	97.0
A	93.0
A-	90.0
B+	87.0
B	83.0
B-	80.0
C+	77.0
C	73.0
C-	70.0
D+	67.0
D	63.0
D-	60.0

Schedule

Week	Date	Day	Planned
1	2-Sep	Tuesday	L1: Introduction to glaciers
	4-Sep	Thursday	L2: Introduction to glaciology
2	9-Sep	Tuesday	L3: Surface mass balance I: accumulations zones
	11-Sep	Thursday	P1: Intro to data analysis in python
3	16-Sep	Tuesday	L4: Mass balance II: ablation
	18-Sep	Thursday	P2: Antarctic climate data
4	23-Sep	Tuesday	L5: Ice as a nonlinear fluid
	25-Sep	Thursday	P3: Intro to solving equations numerically
5	30-Sep	Tuesday	L6: Slow ice flow
	2-Oct	Thursday	P4: Ice sheet modelling I
6	7-Oct	Tuesday	L7: Basal motion.
	9-Oct	Thursday	P5a: Ice sheet modelling II: extending the model
7	14-Oct	Tuesday	-
	16-Oct	Thursday	L8: Ice flow extremes: ice divides and grounding lines
8	21-Oct	Tuesday	P5b: Ice sheet modelling II: extending the model
	23-Oct	Thursday	L9: Ice shelves
9	28-Oct	Tuesday	P6: Computing ice thickness and speed
	30-Oct	Thursday	L10: Heat flow in ice sheets
10	4-Nov	Tuesday	-
	6-Nov	Thursday	P7a: the advection-diffusion equation in ice sheets I
11	11-Nov	Tuesday	L11: Glacial hydrology I
	13-Nov	Thursday	P7b: the advection-diffusion equation in ice sheets II
12	18-Nov	Tuesday	L12: Glacial hydrology II
	20-Nov	Thursday	P7c: the advection-diffusion equation in ice sheets III
13	25-Nov	Tuesday	L13 Ice sheet isochrones and age structure
	27-Nov	Thursday	-
14	2-Dec	Tuesday	P8: Age equation practical
	4-Dec	Thursday	Review and chance to discuss student papers
15	9-Dec	Tuesday	-
	11-Dec	Thursday	-
	16-Dec	Tuesday	class presentations 1
	18-Dec	Thursday	class presentations 2
	19-Dec	Friday	Deadline for final projects

Assignments

Started in practical sessions.

Completed in your own time.

Will require a short write-up, ~1 page, with figures.

Three, spread through the semester:

Antarctic climate “data” (Oct 2nd)

Ice sheet modelling (Nov 6th)

Heat flow modelling (Dec 2nd)

Grade	lower bound
A+	97.0
A	93.0
A-	90.0
B+	87.0
B	83.0
B-	80.0
C+	77.0
C	73.0
C-	70.0
D+	67.0
D	63.0
D-	60.0

Expectations

Expectations of me:

- Explain everything clearly enough for everyone to understand in class or in office hours.
- Care about everyone getting the most out of the course.
- Respond to things in a timely way.

Expectations

Expectations of me:

- Explain everything clearly enough for everyone to understand in class or in office hours.
- Care about everyone getting the most out of the course.
- Respond to things in a timely way.

Expectations of you:

- Be interested.
- Turn in practical work in plenty of time.
- Let me know if you don't understand something.
- Contribute to discussions.

Textbooks

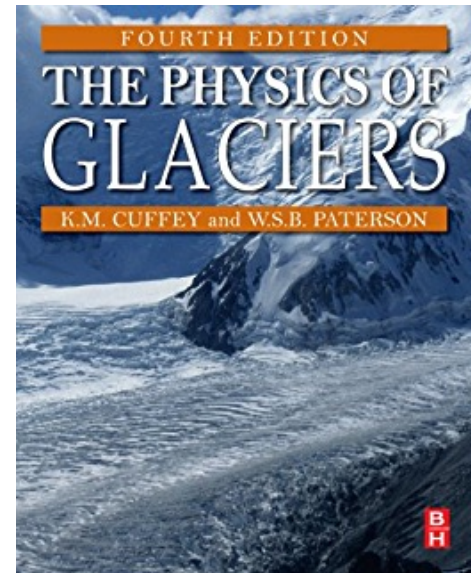
Cuffey, K.M. & W.S.B. Paterson (2010).

The physics of glaciers

Academic Press, 4th Edition.

More mathematical

More references



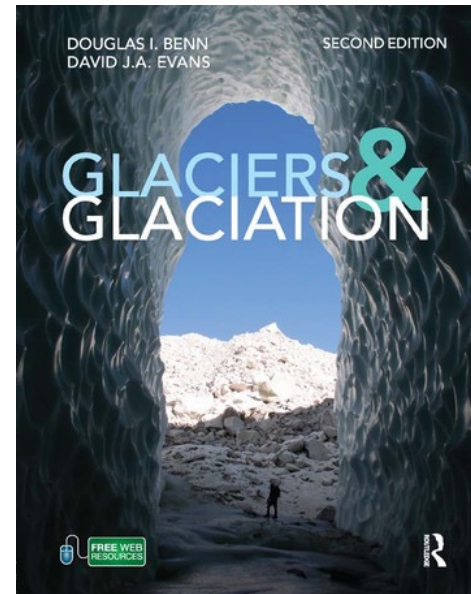
Benn, D. & D. Evans (2010)

Glaciers and Glaciation

Routledge, 2nd Edition.

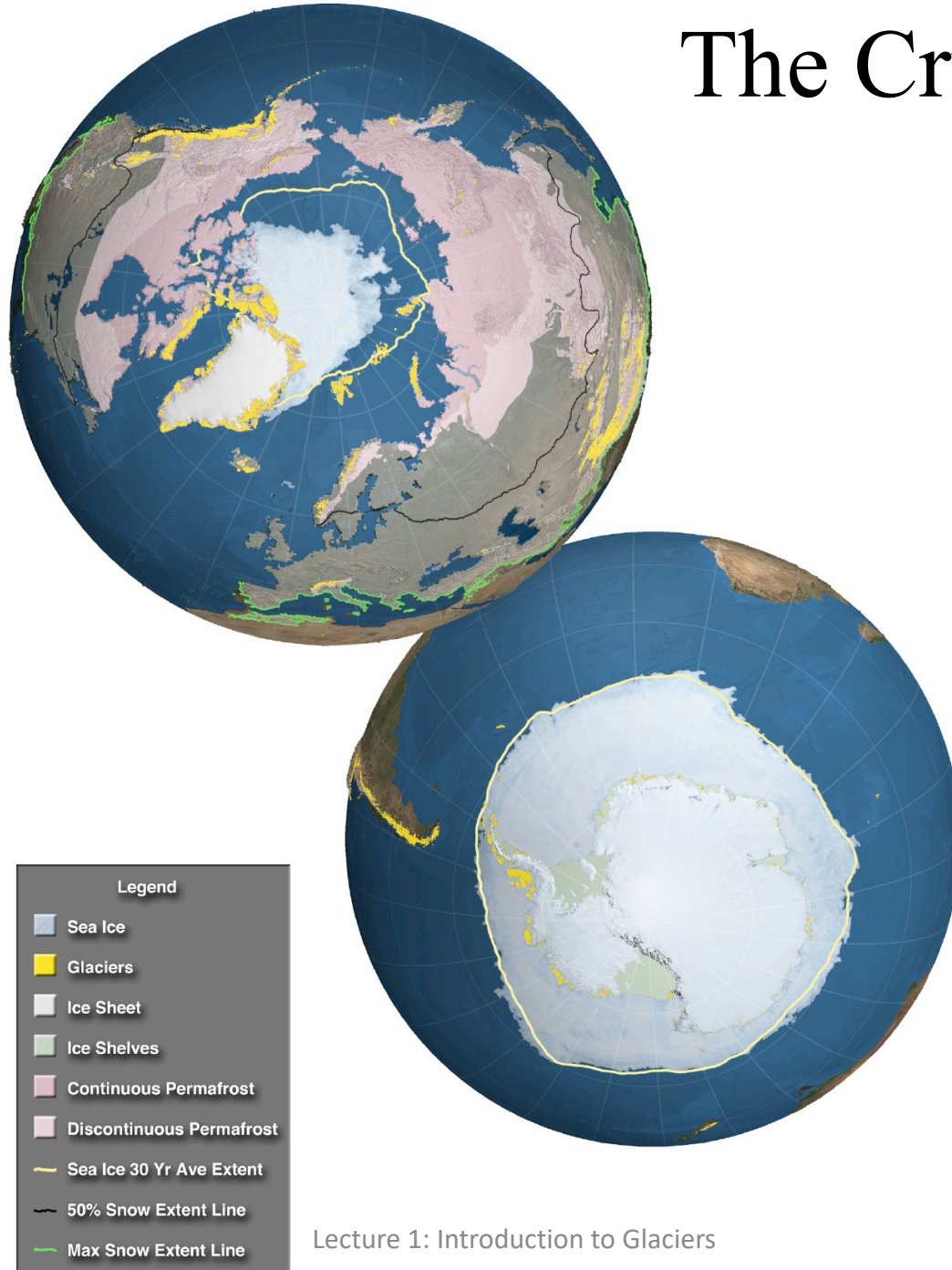
More descriptive

Fewer references



Both out-of-date

The Cryosphere



What are glaciers and ice sheets?

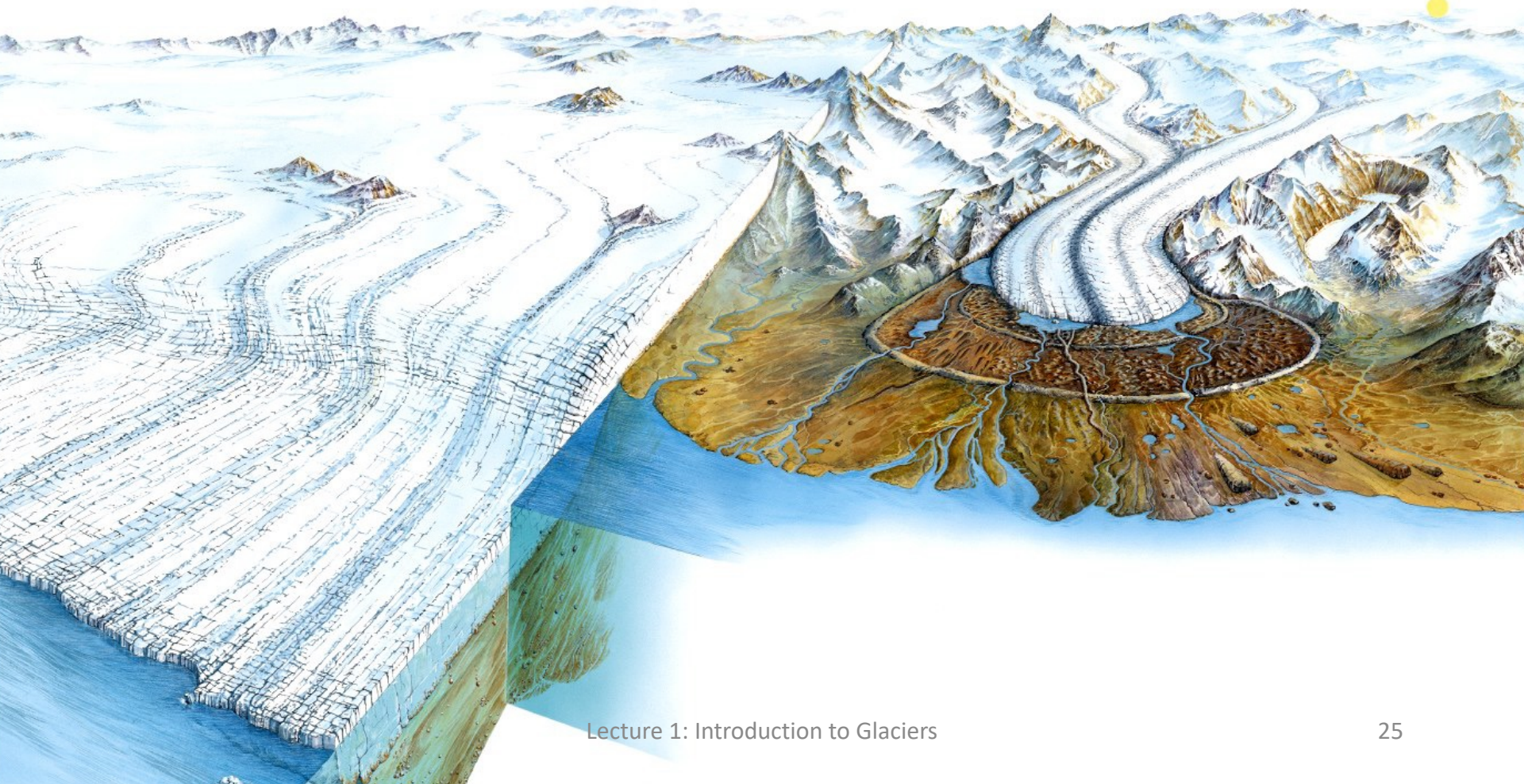
Glaciers and ice sheets

Ice masses that are thick enough to flow under their own weight

10's to 1000's of m thick

Ice sheets

Glaciers





Mer de Glace, France

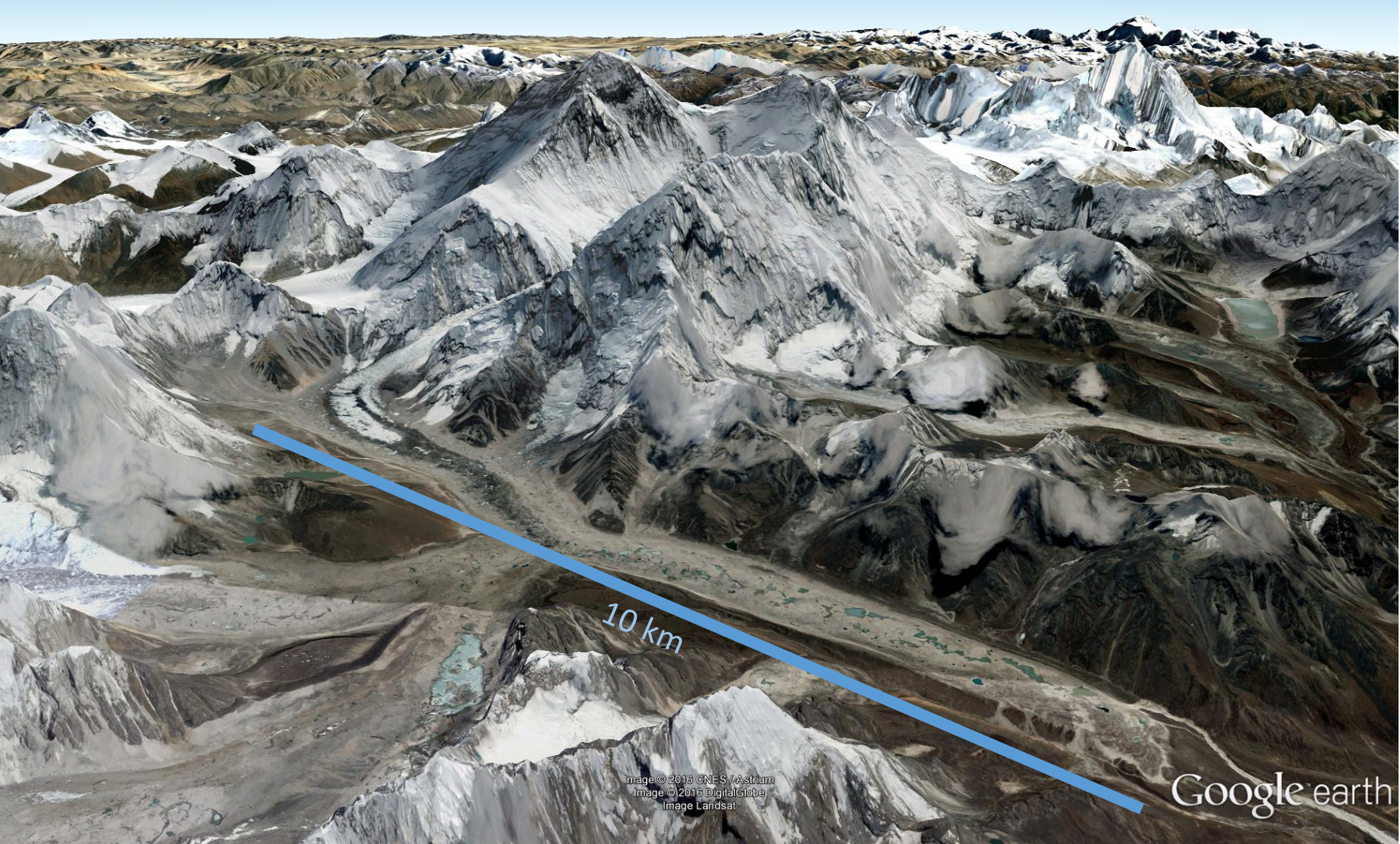
N45.913445° W006.926815° (looking south)



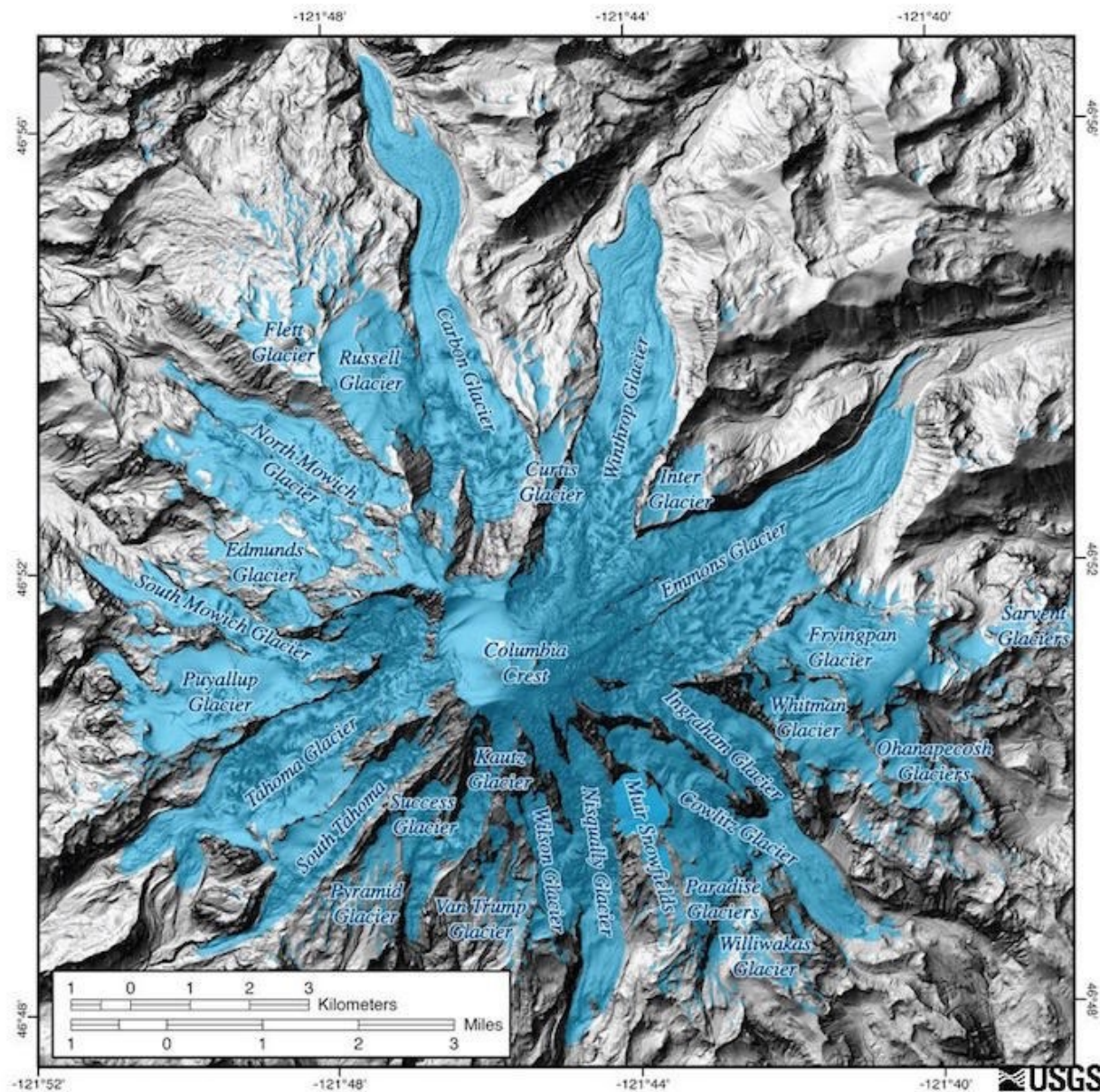


Abandoned tunnels
drilled in glacier for
tourists





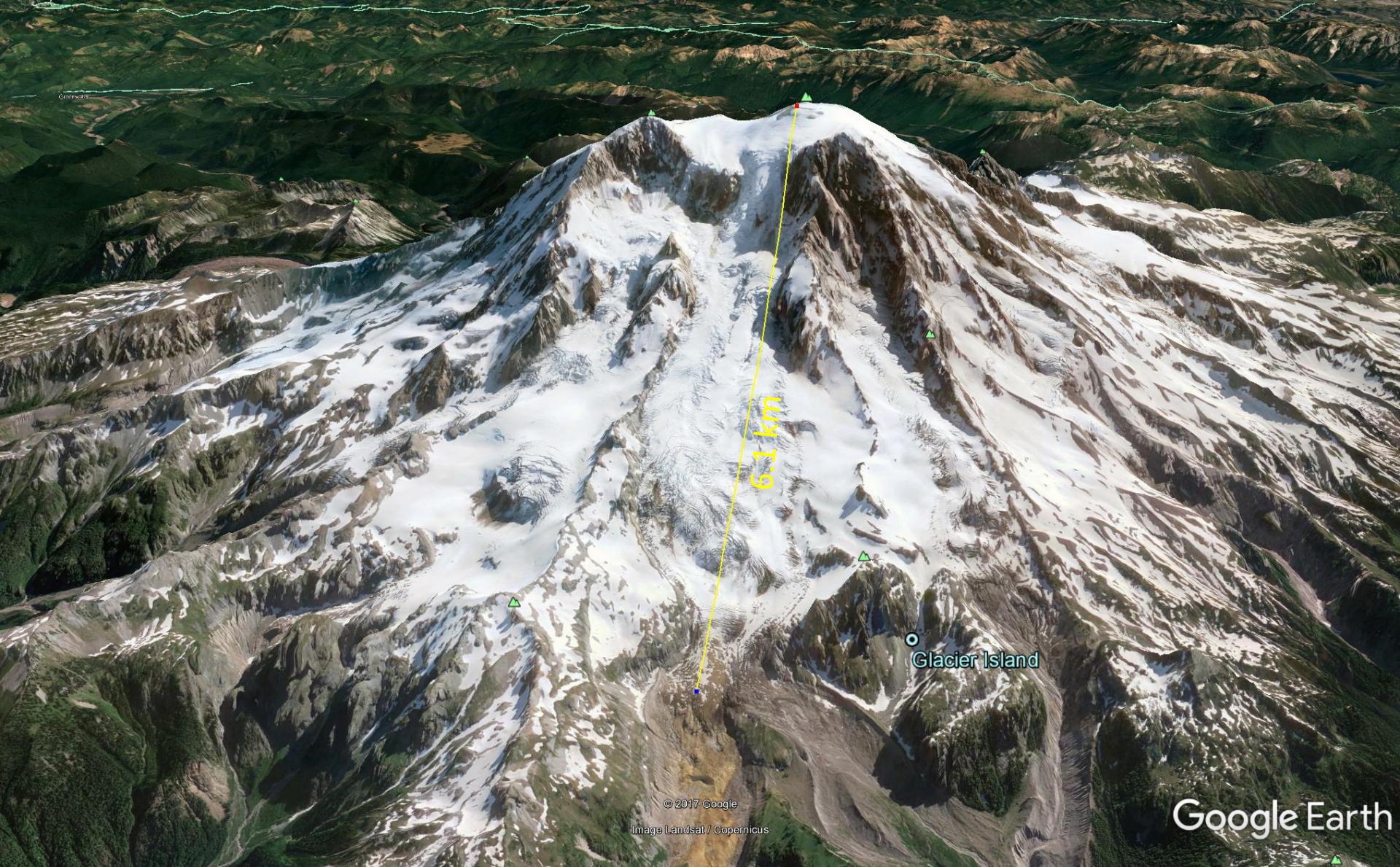
Khumbu Glacier, Nepal
Partly debris-covered mountain glacier.
N27.961978° E086.837274° (looking north)



LIDAR DEM
image from
USGS

Mt. Rainer, Washington State

N46.854203° W121.746943°



Tacoma Glacier,
Washington State

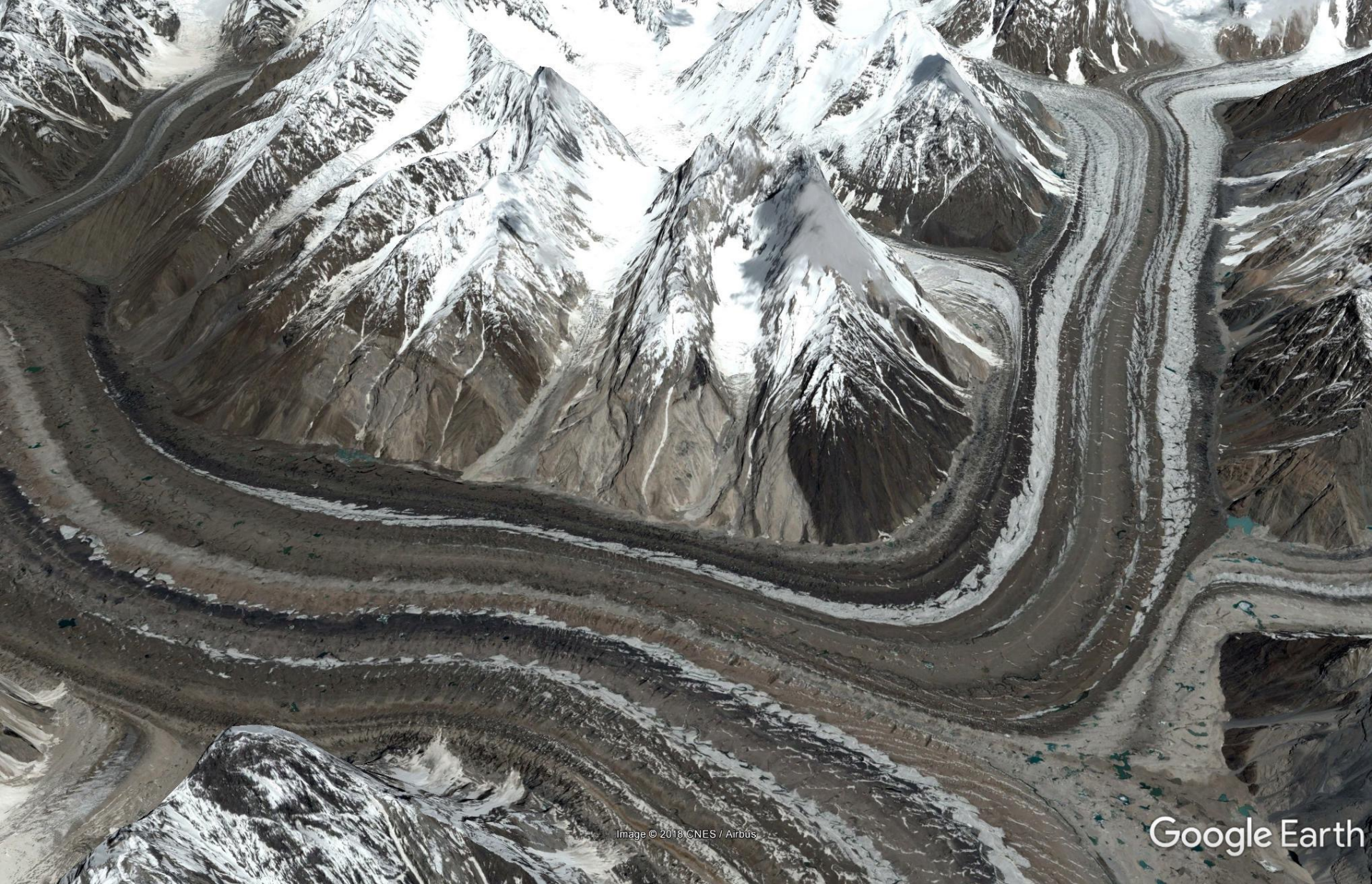
$N46.854203^{\circ}$ $W121.746943^{\circ}$ (looking north east)



Baltoro Glacier, Pakistan

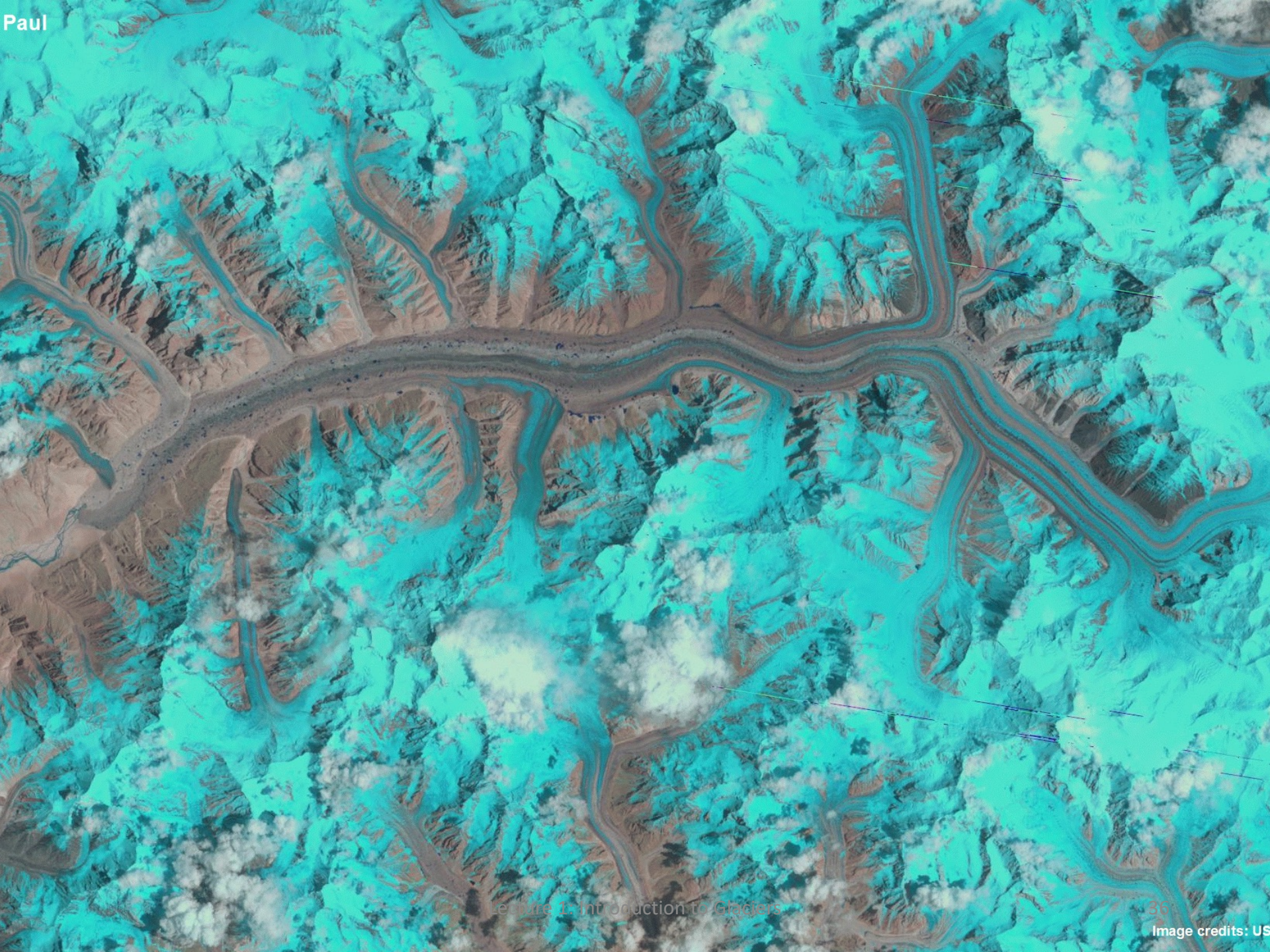
>60km long

N35.694420° E076.376977°



Baltoro Glacier, Pakistan >60km long

N35.694420° E076.376977°



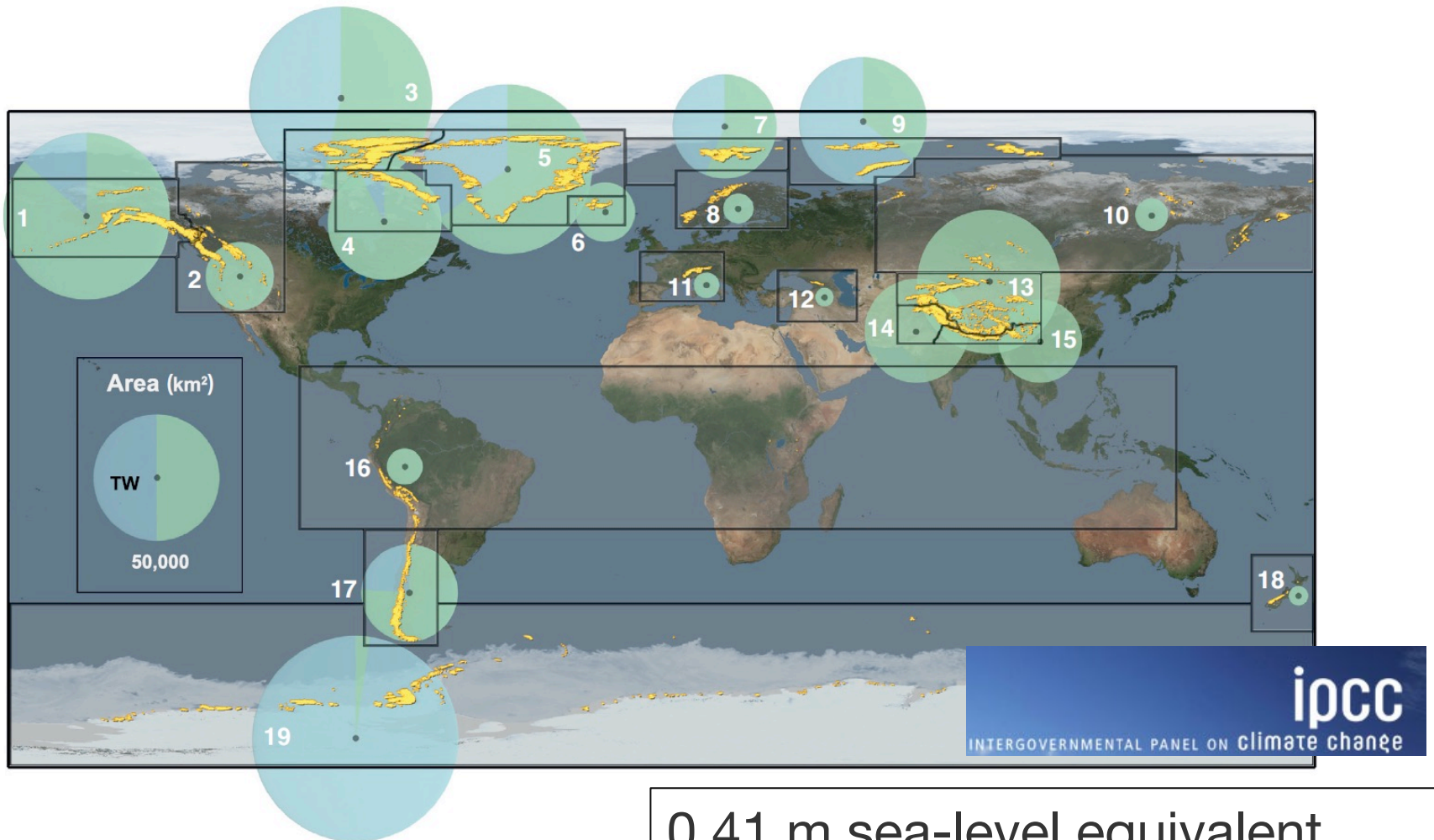
Global distribution of glaciers

Randolph Glacier Inventory 6.0

Region number	Region name	Glacier count	Glacierized area (km ²)
00	World	215547	705738.793
01	Alaska	27108	86725.053
02	Western Canada and US	18855	14524.224
03	Arctic Canada North	4556	105110.642
04	Arctic Canada South	7415	40888.228
05	Greenland Periphery	19306	89717.066
06	Iceland	568	11059.700
07	Svalbard	1615	33958.934
08	Scandinavia	3417	2949.103
09	Russian Arctic	1069	51591.600
10	North Asia	5151	2410.051
11	Central Europe	3927	2092.146
12	Caucasus and Middle East	1888	1306.992
13	Central Asia	54429	49303.415
14	South Asia West	27988	33568.298
15	South Asia East	13119	14734.204
16	Low Latitudes	2939	2341.036
17	Southern Andes	15908	29429.080
18	New Zealand	3537	1161.801
19	Antarctic and Subantarctic	2752	132867.220

RGI Consortium (2017)

Global distribution of glaciers



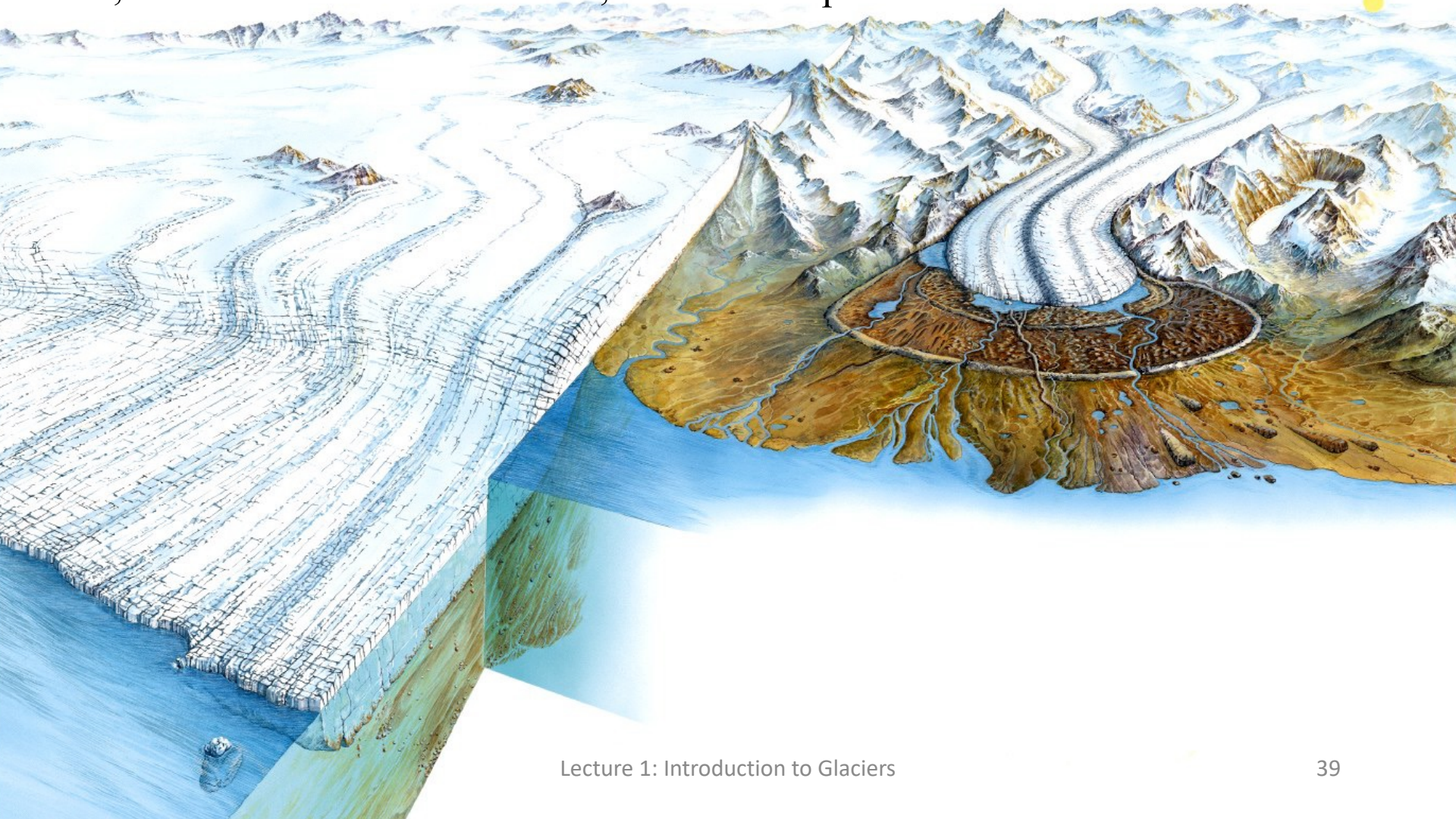
0.41 m sea-level equivalent
IPCC AR5 (Vaughan et al, 2013)

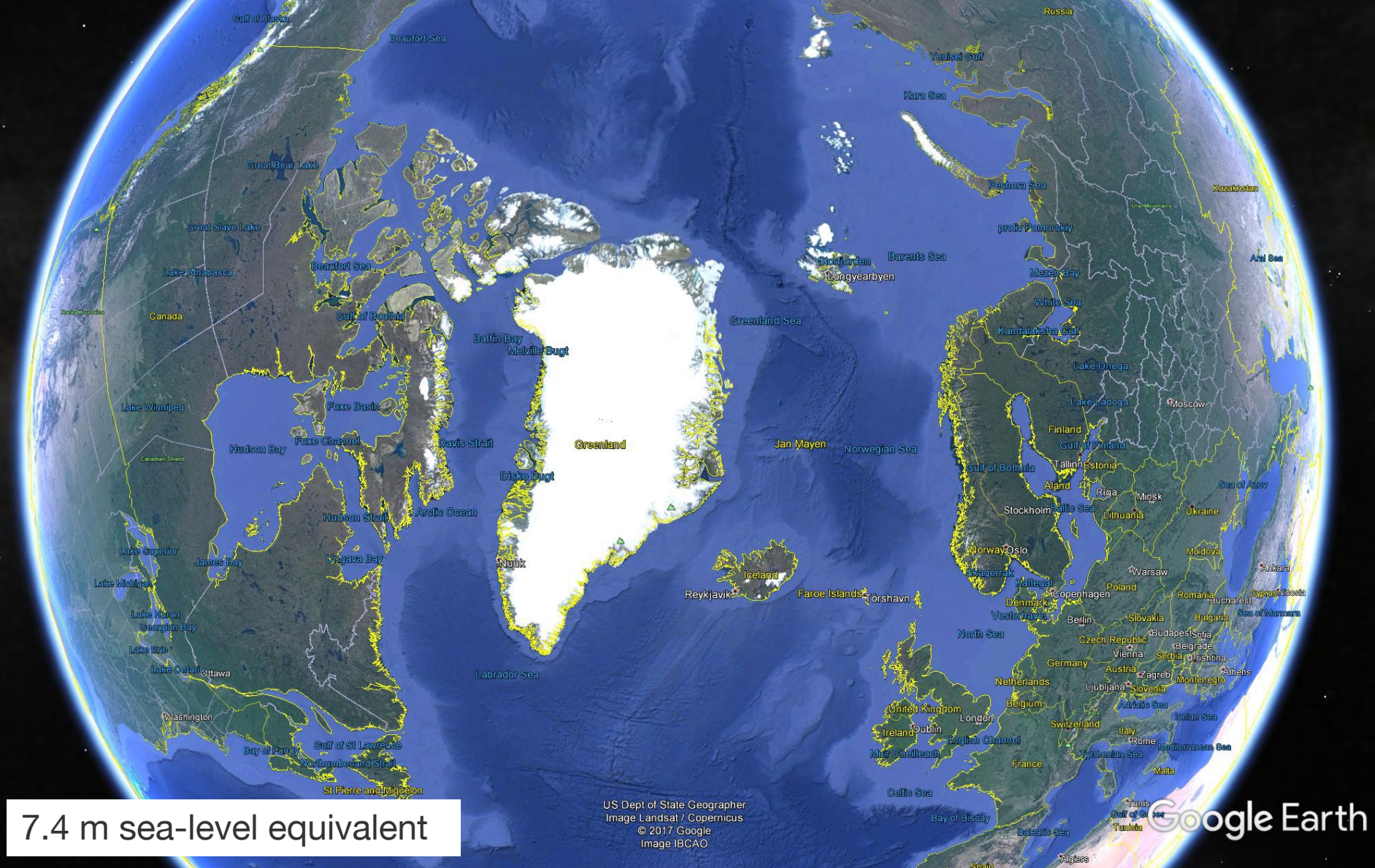
Ice sheets: 100's to 1000's of m thick.

Completely cover terrain

Flow outwards from a 'flow center' towards the 'margin' (ocean or terrestrial).

>50,000 km² makes it an ice sheet, not an ice cap





1.7 million km²

2400 x 1000 km

~ 3000 m thick
Morlighem et al. (2017)



4 million km²

4000-5000 km across

~4800 m thick

Fretwell and Pritchard et al. (2013)

Bed elevation





Topics we will cover

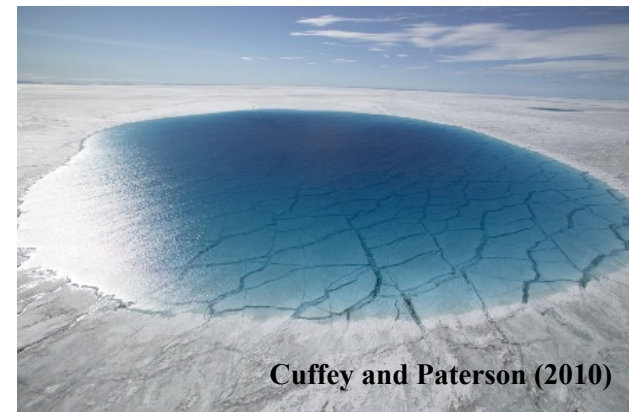
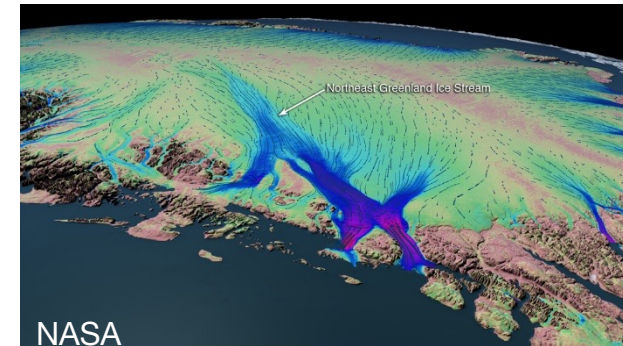
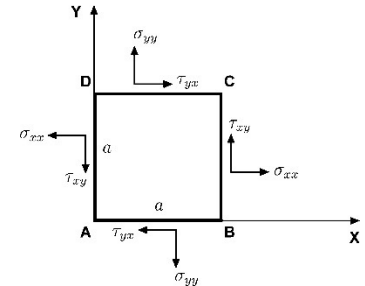
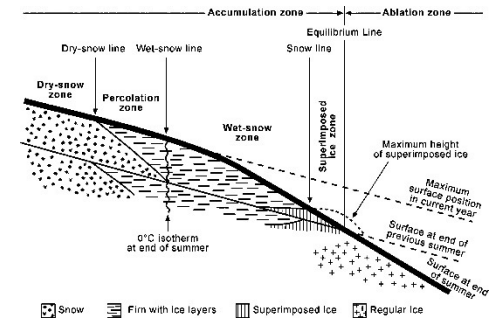
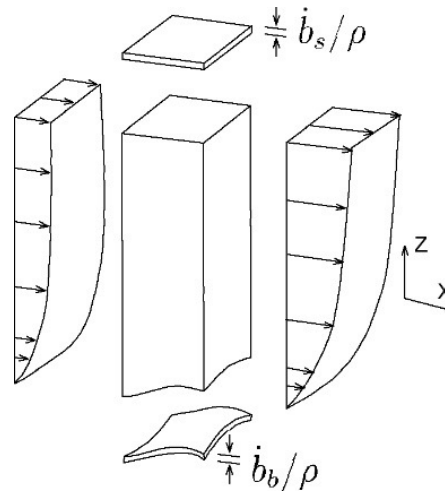
- Accumulation zone processes
- Surface energy balance
- Ablation
- Ice deformation
 - Grain-scale deformation processes
 - Large-scale deformation
- Ice-sheet dynamics
 - Slow flowing areas
 - Ice streams
 - Ice shelves
 - Basal motion

Reading week

- Heat flow
- Glacier hydrology
- The age of ice



Gavin McNamara.



Cuffey and Paterson (2010)

Short reading for Thursday

Cuffey and Paterson

Chapter 1: Introduction (pages 1 – 6)

(scan on Canvas)

Thursday's lecture is at the same time (2:40 pm):

Introduction to glaciology

References

- Benn, D. and Evans, D.J., 2010. *Glaciers and glaciation*. Routledge.
- Cuffey, K.M. and Paterson, W.S.B., 2010. *The physics of glaciers*. Academic Press.
- Fretwell, P., Hamish D. Pritchard, David G. Vaughan, J. L. Bamber, N. E. Barrand, R. Bell, C. Bianchi et al. "Bedmap2: improved ice bed, surface and thickness datasets for Antarctica." *The Cryosphere* 7, no. 1 (Feb., 2013)
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- Plaut, J.J., Picardi, G., Safaieinili, A., Ivanov, A.B., Milkovich, S.M., Cicchetti, A., Kofman, W., Mouginot, J., Farrell, W.M., Phillips, R.J. and Clifford, S.M., 2007. Subsurface radar sounding of the south polar layered deposits of Mars. *science*, 316(5821), pp.92-95.
- RGI Consortium (2017). Randolph Glacier Inventory – A Dataset of Global Glacier Outlines: Version 6.0: Technical Report, Global Land Ice Measurements from Space, Colorado, USA. Digital Media. DOI: <https://doi.org/10.7265/N5-RGI-60>
- Rignot, E., Bamber, J.L., Van Den Broeke, M.R., Davis, C., Li, Y., Van De Berg, W.J. and Van Meijgaard, E., 2008. Recent Antarctic ice mass loss from radar interferometry and regional climate modelling. *Nature geoscience*, 1(2), pp.106-110.
- Tebaldi, C., Strauss, B.H. and Zervas, C.E., 2012. Modelling sea level rise impacts on storm surges along US coasts. *Environmental Research Letters*, 7(1), p.014032. Nye, J.F., 1976. Water flow in glaciers: jökulhlaups, tunnels and veins. *Journal of Glaciology*, 17(76), pp.181-207.
- Vaughan, D.G., J.C. Comiso, I. Allison, J. Carrasco, G. Kaser, R. Kwok, P. Mote, T. Murray, F. Paul, J. Ren, E. Rignot, O. Solomina, K. Steffen and T. Zhang, 2013: Observations: Cryosphere. In: *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* [Stocker, T.F., D. Qin, G.-K. Plattner, M. Tignor, S.K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex and P.M. Midgley (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.
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